Jiahui Li

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University of Georgia, Athens, Georgia, USA
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BIOGRAPHY

Jiahui Li is a Ph.D. student in Computer Science at the University of Georgia, supervised by Dr. Fei Dou. His research focuses on evidence-grounded and explainable artificial intelligence for clinical diagnosis and health sensing, including multimodal clinical reasoning, physiological time-series modeling, and trustworthy evaluation. His work spans electrocardiography, photoplethysmography, ballistocardiography, bodyseismography, vision-language models, and instruction-tuned large language models.

EDUCATION

University of Georgia **Aug. 2024 – Present**
Ph.D. in Computer Science GPA: 4.00/4.00

University College London **2021 – 2022**
M.Sc. in Scientific and Data-Intensive Computing GPA: 3.62/4.00

University of Liverpool **2017 – 2021**
B.Sc. in Computer Science and Engineering GPA: 3.92/4.00

RESEARCH EXPERIENCE

Graduate Research Assistant **Aug. 2024 – Present**
School of Computing, University of Georgia; Advisor: Dr. Fei Dou

- **DynamicDx (Lead):** Built a reproducible benchmark of 71 video-grounded neurological consultations across 11 phenomenologies. Evaluation of five vision-language models localized the primary diagnostic bottleneck to evidence acquisition rather than downstream interpretation.
- **EarlyDx (Lead):** Designed an admission-anchored benchmark for open-ended, evidence-supported emergency-department diagnosis, testing whether generated diagnoses are grounded in retrievable clinical evidence.
- **DeepArrhythmia (Lead):** Developed segment-contextualized electrocardiogram classification with selective acquisition of waveform images, R-peak locations, and rhythm-morphology evidence, producing an auditable decision trace.
- **BEACON-BP (Lead):** Studied contactless blood-pressure estimation from bed-frame geophone sensing using invasive arterial-line measurements as the clinical reference.
- **Peak-Detector / Peak-R1 (Lead):** Built instruction-tuned large-language-model methods for explainable peak detection across electrocardiography, photoplethysmography, ballistocardiography, and bodyseismography without per-modality retraining.
- **ADLGen (Co-Lead, co-first author):** Synthesized symbolic, event-triggered smart-home sensor sequences for activity modeling and ambient assisted living.

Research Assistant **2023 – 2024**
Hong Kong University of Science and Technology; Supervisor: Prof. Dit-Yan Yeung

- Built **MGTST**, a multi-scale, cross-channel gated Transformer for multivariate long-term forecasting in high-dimensional time series.

PUBLICATIONS

Full publication details are available on Google Scholar. * indicates equal contribution, † indicates correspondence.

*********First Author Papers*********

- [**Submitted**] **Jiahui Li**, Yutong Guo, Nan Yang, Wenzhan Song, Jin Lu, Fei Dou†. "DynamicDx: Evaluating Perception and Evidence Acquisition in Video-Grounded Conversational Diagnosis." Under review at **ICLR 2027**.
- [**Submitted**] **Jiahui Li**, Ruili Fang, Zishuai Liu, Yutong Guo, Nan Yang, Wenzhan Song, Jin Lu, Fei Dou†. "EarlyDx: An Admission-Anchored Benchmark for Open-Ended Generation of Evidence-Supported ED-Encounter Diagnoses." Under review at **KDD, 2027**. [link]
- [**Submitted**] **Jiahui Li**, Yuanyuan Zhang, Yida Zhang, Zixuan Zeng, Jiayu Chen, Yingjian Song, Yin Xiao, Nishan Dong, Junjie Lu, Xiang Zhang, Jin Lu, Wenzhan Song, Fei Dou†. "BEACON-BP: A Feasibility Study of Bed-Frame Geophone Sensing for Contactless Blood Pressure Estimation with Arterial-Line Reference." Under review at **IMWUT, 2026**.

- [**IMWUT'26**] **Jiahui Li**, Yida Zhang, Zixuan Zeng, Jiayu Chen, Yingjian Song, Yin Xiao, Nishan Dong, Junjie Lu, Younghoon Kwon, Xiang Zhang, Jin Lu, Wenzhan Song, Fei Dou†. "Peak-Detector: Explainable Peak Detection via Instruction-Tuned Large Language Models in Physiological Signal." Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, **IMWUT (UbiComp) 2026, CSRankings.** [link]
- [**SenSys'26**] Weihang You*, Hanqi Jiang*, **Jiahui Li*** **(Co-first author)**, Zishuai Liu*, Tianming Liu, Jin Lu, Fei Dou†. "ADLGen: Synthesizing Symbolic, Event-Triggered Sensor Sequences for Smart-Home Human Activity Modeling." Proceedings of the 2026 ACM/IEEE International Conference on Embedded Artificial Intelligence and Sensing Systems, **SenSys 2026, CORE A*/CSRankings**. [link]
- [**NeurIPS'26**] **Jiahui Li**, Ruili Fang, Zishuai Liu, Wenzhan Song, Jin Lu, Fei Dou†. "DeepArrhythmia: Segment-Contextualized ECG Arrhythmia Classification via Selective Evidence Acquisition." To appear at **NeurIPS 2026, CORE A*/CSRankings**. [link]
- [**TS4H@NeurIPS'25**] **Jiahui Li**, Yida Zhang, Zixuan Zeng, Jiayu Chen, Xiang Zhang, Jin Lu, Wenzhan Song, Fei Dou†. "Peak-R1: Instruction-Tuned Large Language Models for Robust J-Peak Detection." Learning from Time Series for Health Workshop at the 39th Annual Conference on Neural Information Processing Systems, 2025 [link]

*********Collaborative Author Papers*********

- [**Preprint**] Yuanyuan Zhang*, Yuyan Wu*, Sahas Biddappa, Yida Zhang, Jiayu Chen, Zixuan Zeng, Yixuan Li, **Jiahui Li**, Fei Dou, Zhenlin An, Younghoon Kwon, Hae Young Noh, WenZhan Song. "Bodyseismography (BSG): Theoretical Foundation and Numerical Modeling." **arXiv preprint, 2026.** [link]
- [**HumanSys@MobiCom'26**] Yuyan Wu, Yuanyuan Zhang, Yida Zhang, **Jiahui Li**, Fei Dou, Wenzhan Song, Hae Young Noh. "Contact-Free Continuous ABP Waveform Reconstruction via Triaxial Bodyseismography." The 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems, 2026.
- [**TMC'26**] Yuanyuan Zhang, Yida Zhang, **Jiahui Li**, Yuyan Wu, Fei Dou, Xiao Yin, Zhenlin An, Hae Young Noh, Wenzhan Song. "Physics-Constrained Deep Learning Model for Contactless Blood Pressure Monitoring from Triaxial Bodyseismography." Major Revision at **IEEE Transactions on Mobile Computing**. [link]
- [**TIP'26**] Md Shadab Alam, Linghan Zhang, **Jiahui Li**, Fei Dou, Pavlo Bazilinskyy. "Collision Patterns and Reporting Blind Spots in 971 California Autonomous Vehicle Crash Reports." Major Revision at **Traffic Injury Prevention**.
- [**Submitted**] Yida Zhang, Zixuan Zeng, Jiayu Chen, Yuanyuan Zhang, Yingjian Song, **Jiahui Li**, Xiang Zhang, Zaipeng Xie, Xiao Yin, Jie Yuan, Fei Dou, Wenzhan Song. "SeismoAF: AFib Detection from BodySeismoGraphy via Physiology-Guided AFib Synthesis and Context-Aware Adaptation." Under review at **IMWUT 2026**.
- [**Submitted**] Yida Zhang, Yuanyuan Zhang, Jiayu Chen, Zixuan Zeng, Yingjian Song, **Jiahui Li**, Xiao Yin, Zaipeng Xie, Xiang Zhang, Guoliang Xing, Zhenlin An, Fei Dou, Wenzhan Song. "BSG2ECG: Contactless Electrocardiogram Monitoring with Triaxial Bodyseismography." Under review at **PerCom 2027**.
- [**Submitted**] Jiayu Chen, Zixuan Zeng, Yingjian Song, Yida Zhang, **Jiahui Li**, Zhenlin An, Xiang Zhang, Fei Dou, Deepak Das, Bradley Phillips, Wenzhan Song. "SeismoApnea: Bodyseismography-Based Overnight Sleep Apnea Monitoring and Apnea-Hypopnea Index Estimation." Under review at **PerCom, 2027**.
- [**Submitted**] Zixuan Zeng, Jiayu Chen, Yida Zhang, Yingjian Song, **Jiahui Li**, Zaid Pitafi, Xiang Zhang, Fei Dou, Deepak Das, Bradley Phillips, Wenzhan Song. "BSGSleep: Sleep Quality Monitoring via Multi-Axis Bodyseismography." Under review at **PerCom 2027**.
- [**Preprint**] Haoran Lu, Luyang Fang, Ruidong Zhang, and others, including **Jiahui Li**. "Alignment and Safety in Large Language Models: Safety Mechanisms, Training Paradigms, and Emerging Challenges." **arXiv preprint, 2025**. [link]
- [**Preprint**] Zicheng Liu, **Jiahui Li**, Siyuan Li, Zelin Zang, Cheng Tan, Yufei Huang, Yajing Bai, Stan Z. Li. "GenBench: A Benchmarking Suite for Systematic Evaluation of Genomic Foundation Models." **arXiv preprint, 2024**. [link]

TEACHING ASSISTANT

CSCI 1302: Software Development, Fall 2024 and Spring 2025, University of Georgia
CSCI 4470: Algorithms, Fall 2025 and Fall 2026, University of Georgia

HONORS AND AWARDS

UGA James L. Carmon Scholarship Award (2027)
— A university-wide graduate research award with one primary recipient selected annually.

NeurIPS 2026 Financial Assistance Award
— Full conference and workshop registration, plus an Atlanta hotel stay.